

Supporting Information

Structure-Aware Row Scaling for Block-Structured Mixed-Integer Programs

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This document contains (i) the 0.1%-MIPGap Gurobi distributional, diagnostic, and performance-profile views (Scenario 2 of the main text; Figures S1–S3), which mirror the corresponding 1%-MIPGap figures of the main text; (ii) the 0.1%-MIPGap CPLEX distributional and profile views (Scenario 4; Figure S4); and (iii) the hyperparameter sensitivity of the structure-aware rule (Table S1). The quantitative evidence for every scenario is in the main-text tables.

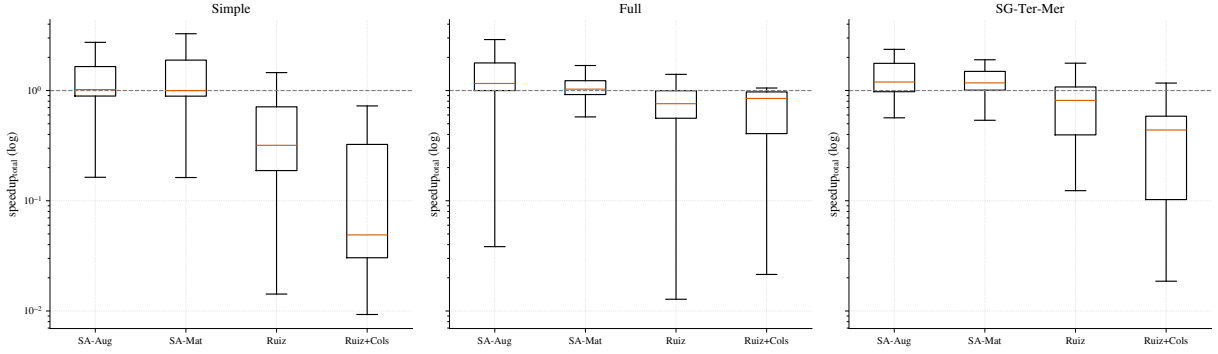


Figure S1: At 0.1% MIPGap, the Gurobi speedup distributions preserve the same qualitative pattern observed at the looser tolerance. Values above one indicate faster runs than the unscaled formulation.

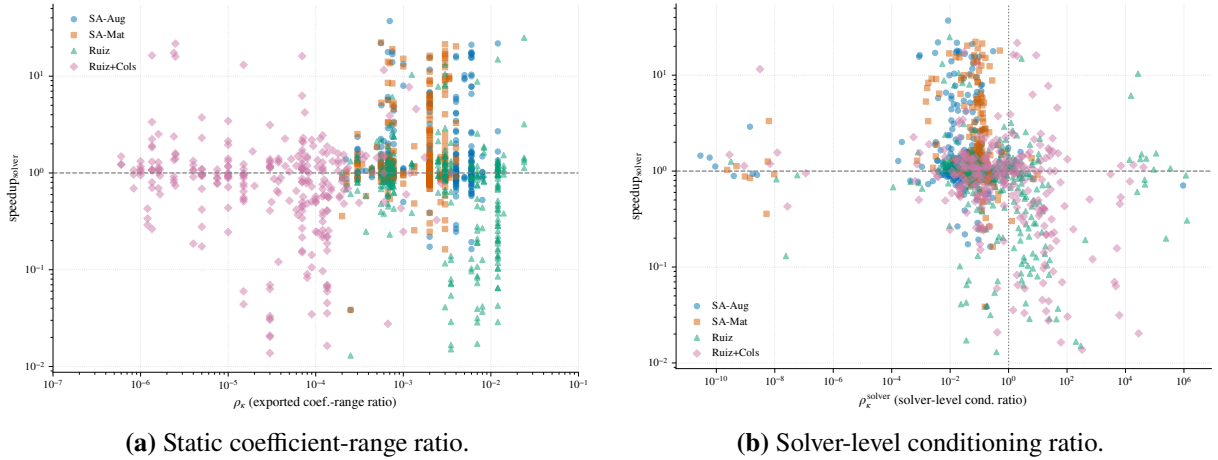


Figure S2: At 0.1% MIPGap, the stricter Gurobi runs again separate static coefficient compression from solver-facing numerical behavior. The solver-level diagnostic remains more informative than the exported coefficient-range ratio.

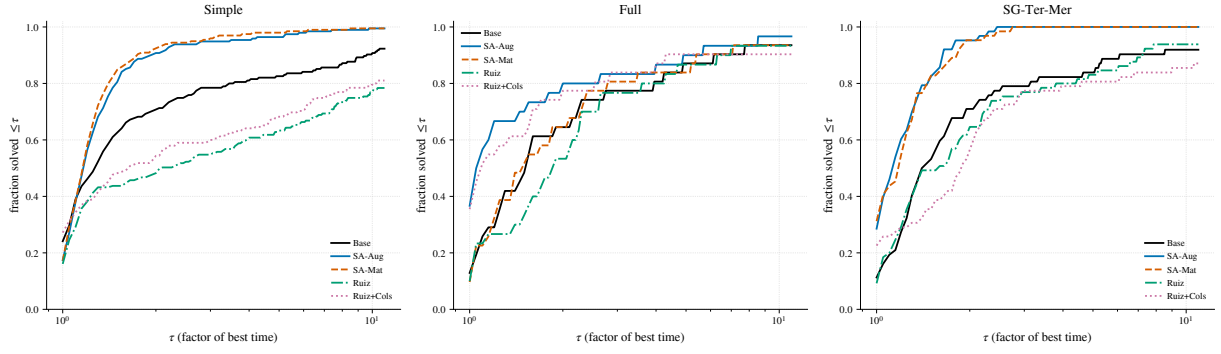


Figure S3: At 0.1% MIPGap, the Dolan–Moré profiles test whether the Gurobi speedups persist across the heterogeneous instance pool rather than only in averages.

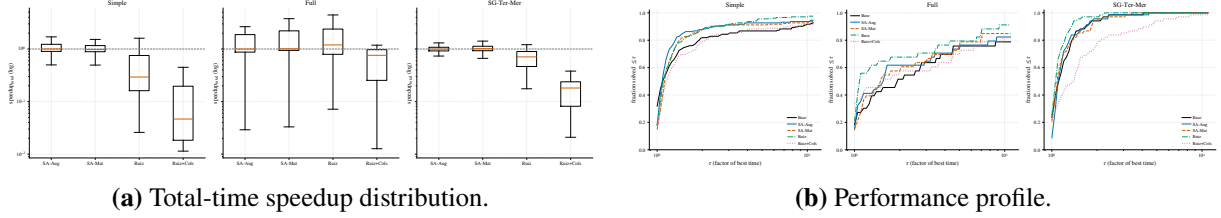


Figure S4: At 0.1% MIPGap, the CPLEX visual evidence remains mixed under the stricter tolerance, supporting the solver-dependent interpretation of the main text.

Table S1: Hyperparameter sensitivity of SA-Aug (Simple class, Gurobi, 1% MIPGap). Solver-time sgm ($s = 1$ s) and PAR10, in seconds; default settings in bold.

Parameter	Value	sgm (s)	PAR10 (s)
Band u	1.5	2.09	2.46
	2	2.07	2.45
	4	2.12	2.49
Window $[\varepsilon_{\min}, \varepsilon_{\max}]$	$[10^{-6}, 10^6]$	2.04	2.40
	$[10^{-9}, 10^9]$	2.02	2.39
	$[10^{-12}, 10^{12}]$	2.01	2.36

Base sgm is 3.73 s for reference. Each row is a separate matched run; the spread across all settings ($\text{sgm} \in [2.01, 2.12]$) is far below the base-to-scaled gap, and the median paired speedup is ≈ 1.0 throughout. The number of Ruiz iterations K is a parameter of the Ruiz baselines, not of the structure-aware rule, so it is fixed at $K = 4$; varying it leaves SA-Aug unchanged, as expected.